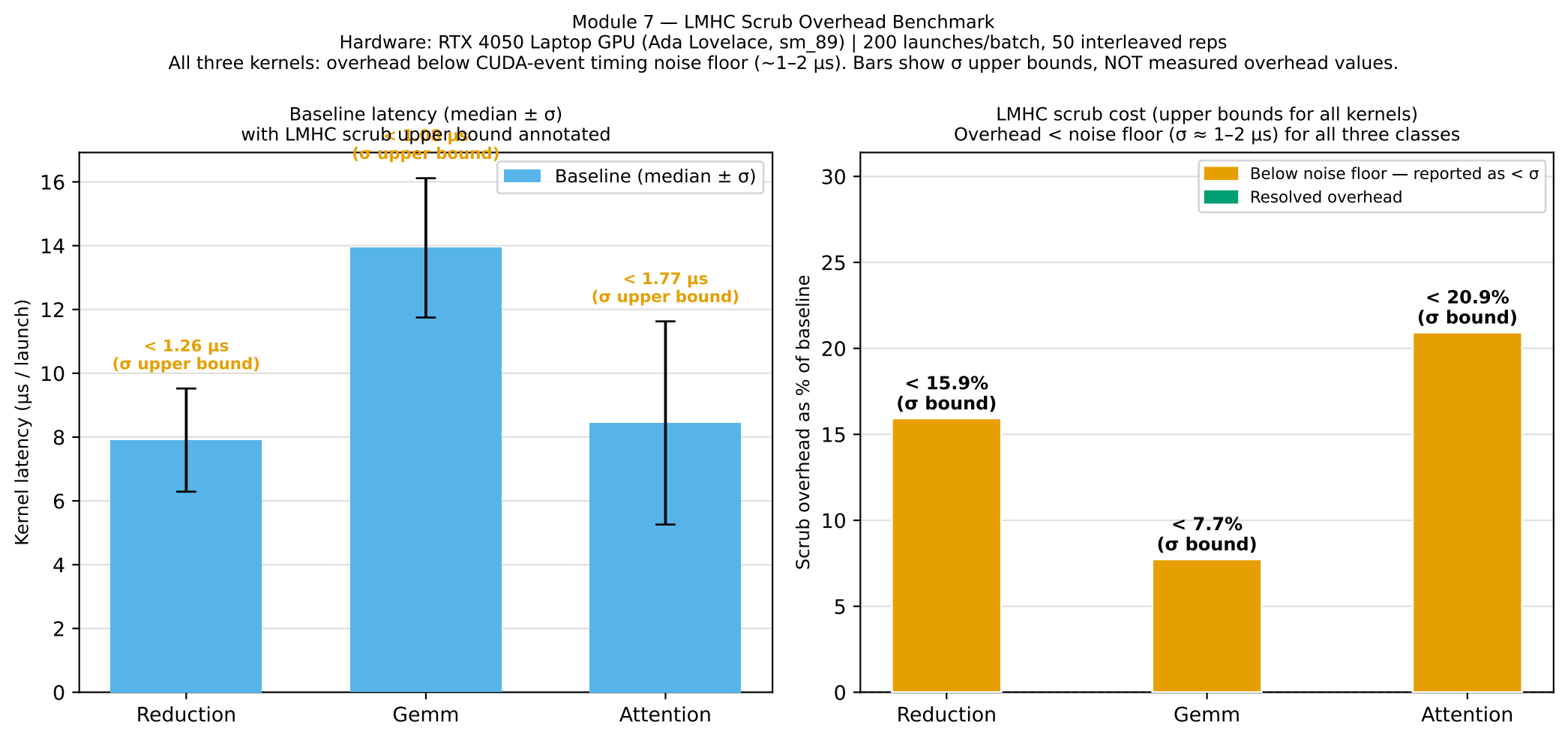
Module 7 — LMHC Scrub Overhead Benchmark
Hardware: RTX 4050 Laptop GPU (Ada Lovelace, sm_89) | 200 launches/batch, 50 interleaved reps
All three kernels: overhead below CUDA-event timing noise floor (~1–2 µs). Bars show σ upper bounds, NOT measured overhead values.
Baseline latency (median ± σ)
with LMHC scrub upper bound annotated
Kernel latency (µs / launch)
Baseline (median ± σ)
< 1.26 µs
(σ upper bound)
< 2.00 µs
(σ upper bound)
< 1.77 µs
(σ upper bound)
Reduction
Gemm
Attention
LMHC scrub cost (upper bounds for all kernels)
Overhead < noise floor (σ ≈ 1–2 µs) for all three classes
Scrub overhead as % of baseline
Below noise floor — reported as < σ
Resolved overhead
< 15.9%
(σ bound)
< 7.7%
(σ bound)
< 20.9%
(σ bound)
Reduction
Gemm
Attention